

Lecture: Dynamics Review

Optimization and Learning for Robot Control

RCI Lab
Dept. of Mechanical Engineering

Kyung Hee University

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1. Robot Dynamic Modeling: Overview

Robot modeling encompasses the mathematical description of motion, forces, and energy.

- **Kinematics (Forward)**: Mapping joint variables to end-effector pose.
- **Statics**: Relating end-effector wrenches to joint torques.
- **Dynamics**: Study of the relationship between forces and resulting motion.

Core Equation of Motion:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau + J^T(q)w$$

2. Robot Manipulator Basics

A robot manipulator is a set of rigid bodies (links) connected by joints.

- **Links and Joints:** Typically joints are either **revolute** or **prismatic**.
- **Actuation:** Joints are driven by motors and transmission systems (e.g., gearboxes).
- **Coordinates:** q_i represents the i -th joint coordinate. The state is defined by the vector $q = [q_1, q_2, \dots, q_n]^T$.
- **End-Effector:** The tool or hand at the end of the kinematic chain.

3. Homogeneous Transformation and Pose

The **pose** (position and orientation) of a frame is defined using a 4×4 transformation matrix:

$$H_1^0 = \begin{bmatrix} R_1^0 & P \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}$$

- **Position Vector (P):** The translation from the origin.
- **Rotation Matrix (R_1^0):** Composed of unit vectors $[x_1^0, y_1^0, z_1^0]$.
- **Orthogonality:** R is orthogonal, meaning $R^{-1} = R^T$.
- **Composition:** For n joints, $H_n^0(q) = H_1^0(q_1)H_2^1(q_2) \dots H_n^{n-1}(q_n)$.

4. Representation of Orientation

Various methods exist to describe the orientation part of the pose:

- **Rotation Matrix:** 3×3 matrix with 9 elements.
- **Minimal Representations:** Euler Angles and Roll-Pitch-Yaw (3 parameters).
 - *Warning:* These suffer from **Singularities**.
- **Non-Minimal Representations:** Angle-axis and **Quaternions** (4 parameters).

5. Differential Kinematics: Geometric Jacobian

We define the relationship between joint velocities and end-effector velocity V_e :

$$V_e = \begin{bmatrix} \dot{p} \\ \omega \end{bmatrix} = \begin{bmatrix} J_p \\ J_o \end{bmatrix} \dot{q} = J(q) \dot{q}$$

- **Linear Velocity (J_p):** Sum of contributions from each joint.
- **Angular Velocity (J_o):** Based on the rotation axes of the joints.
- **Analytical Jacobian (J_A):** Relates $\dot{q}$ to the derivative of pose coordinates $\dot{\phi}$.
- **Note:** $J_A \neq J$ in general for orientation coordinates.

6. Statics: Mapping Forces to Torques

Given a wrench $\tilde{w} = [f^T, m^T]^T$ at the end-effector, we compute joint torques τ .

Power Conservation Principle:

- Power at joints: $P_\tau = \tau^T \dot{q}$.
- Power at end-effector: $P_e = f^T \dot{p} + m^T \omega = w^T V$.

Duality Derivation:

$$\tau^T \dot{q} = w^T J \dot{q} \implies \tau = J^T(q)w$$

This fundamental relation is used to handle external contact forces in control.

7. Properties of the Mass Matrix $M(q)$

The mass matrix represents the inertia of the robot in joint space.

- **Symmetry:** $M(q) = M(q)^T$.
- **Positive-Definiteness:** $M(q) \succ 0$. Kinetic energy $T = \frac{1}{2} \dot{q}^T M(q) \dot{q}$ is always positive for any non-zero $\dot{q}$.
- **Boundedness:** There exist positive constants m_{min} and m_{max} such that:

$$m_{min} I \leq M(q) \leq m_{max} I$$

- **Configuration Dependency:** M is a function of q , representing how inertia changes as the robot's posture changes.

8. Properties of Coriolis and Centrifugal Forces

The term $C(q, \dot{q})\dot{q}$ (or $h(q, \dot{q})$) accounts for velocity-dependent forces.

Skew-Symmetry Property: The matrix $N(q, \dot{q}) = \dot{M}(q) - 2C(q, \dot{q})$ is **skew-symmetric**.

$$\dot{q}^T (\dot{M} - 2C) \dot{q} = 0$$

- *Physical Significance:* This property reflects the conservation of energy in the system.
- *Control Application:* It is a critical component in proving the stability of adaptive and non-linear controllers (e.g., Slotine-Li).

9. Example: 2-DOF Planar RR Robot

For a 2-link robot with masses m_i concentrated at the end of links l_i : **Mass Matrix Components:**

- $M_{11} = (m_1 + m_2)l_1^2 + m_2l_2^2 + 2m_2l_1l_2 \cos(q_2)$
- $M_{12} = M_{21} = m_2l_2^2 + m_2l_1l_2 \cos(q_2)$
- $M_{22} = m_2l_2^2$

Coriolis Matrix Example:

$$C(q, \dot{q}) = \begin{bmatrix} -m_2l_1l_2 \sin(q_2)\dot{q}_2 & -m_2l_1l_2 \sin(q_2)(\dot{q}_1 + \dot{q}_2) \\ m_2l_1l_2 \sin(q_2)\dot{q}_1 & 0 \end{bmatrix}$$

Notice the coupling between joints through the $\sin(q_2)$ term.

10. System Identification: Linearity in Parameters

A key property for robot learning is that the dynamics are **linear** with respect to the inertial parameters π :

$$\tau = Y(q, \dot{q}, \ddot{q})\pi$$

- $\pi \in \mathbb{R}^p$: Vector of inertial parameters (mass, center of mass, inertia moments).
- $Y \in \mathbb{R}^{n \times p}$: The **Regressor Matrix**, which is a function of the measured state $(q, \dot{q}, \ddot{q})$ only.

Parameter Estimation (Least Squares): To "learn" the model, we minimize the error $\|\mathcal{T} - \mathcal{Y}\pi\|^2$:

$$\hat{\pi} = (\mathcal{Y}^T \mathcal{Y})^{-1} \mathcal{Y}^T \mathcal{T}$$

where $\mathcal{T}$ and $\mathcal{Y}$ are stacked measurements over time.